Rafi Irtiza Rasa

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SUMMARY

Software engineer and AI-orchestration lead. 6+ years owning production systems end-to-end for an international organization — then I rebuilt my own legacy: one modern analytics platform replaced the four desktop dashboards I'd hand-built, with a golden validation proving 0 mismatches against the old engine. I direct AI agents as an implementation team, shipping production platforms in days-to-weeks with heavy automated-test coverage, and I built my own orchestration infrastructure (session tracking, LLM cost accounting, agent workflows) to do it. This year I shipped a governed LLM feature into that same production platform — grounded answers, a PII boundary, kill switches, budgets, and an eval suite that gates releases. 17 of my applications are live in production today. Published researcher.

TECHNICAL SKILLS

AI-Augmented Engineering: AI-agent orchestration (Claude Code — multi-agent workflows, spec-driven direction, verification gates), production LLM governance (single-gateway access, type-level PII boundary, kill switches, token budgets, audit/spend ledger, RBAC entitlements), LLM eval harnesses with prompt-injection and refusal gates, RAG pipelines (Qdrant, ChromaDB), verification harnesses for AI-written code (golden validations, parity checkers)
Languages: Python, JavaScript, TypeScript, PHP, Kotlin, Java, C#, SQL
Frontend: React 18/19, Next.js 14–16, Vue 3, Tailwind CSS, Vite
Backend: FastAPI, Laravel 11–13, Next.js server actions, Custom PHP MVC, .NET 4.8
Mobile: Android SDK, Jetpack Compose, Retrofit, Firebase
Database: PostgreSQL, MySQL, SQLite, Redis, Qdrant, ChromaDB
Testing: PHPUnit, Playwright, Vitest, JUnit, MockK, Selenium
Security: RBAC (115+ permissions), threat detection & auto-blocking, JWT, Sanctum
DevOps: Nginx, Docker, Git, PM2, Linux/VPS, Prometheus/Grafana

EXPERIENCE

Software Engineer — Max Foundation (international organization, Netherlands HQ) | Dhaka | Nov 2019 – Present
Sole engineer for the entire Bangladesh office — full ownership of all software across 4 programs and 8+ partner organizations. 125+ end users.

- Built and maintain the entire software ecosystem — 5 applications (Android, C# desktop, PHP web), 1M+ data points across 4 programs, used by 8+ partner organizations
- Designed a form generation system driven by database configuration — field teams deploy new surveys without code changes, replacing what used to be a 2-week recompilation cycle
- Replaced my own legacy: one Next.js analytics platform superseded the 4 desktop dashboards I'd built years earlier — a faithful port of the WHO z-score engine, validated against 5,000 golden samples with zero mismatches — and it has since gone from 2 field programs to 6
- Put a governed AI query layer on that platform: natural-language questions get answered by running the dashboard's own charts, so the model never authors a number — behind a type-level PII boundary, deny-by-default kill switches, per-module entitlements, and an audit ledger that doubles as the spend ledger, all gated by an eval suite with prompt-injection and refusal checks
- Rebuilt the organization's decade-old PHP HR platform in Next.js/TypeScript and ran the cutover myself, reconciling every migrated table row-for-row against the legacy database (30 staff, 281 leave requests, 10,143 attendance rows) before the flip — it is now the live system of record
- Took over a vendor-built ERP (Laravel + Flutter + admin, 52 models, 85 migrations), retrofitted ~56 architecture docs, shipped new features, and ran the staged production cutover

Technical Consultant — Funded Academy → Titan Development | Dhaka (remote) | Sep 2025 – Present

Retained by one owner across two companies: ran the full technology stack of a prop-trading platform (Funded Academy, fintech, Sep 2025 – Sep 2026), now web development and maintenance for a New York construction firm (Titan).

- Integrated the MetaTrader 5 Web API over a raw TLS socket — reverse-engineered the connection-bound auth protocol after HTTP clients proved unusable, verified against the live server
- Built a live trading-competition engine that polls MT5 group equity in batch, snapshots ticks, and recomputes rankings — spectators never touch the trade server
- Architected and shipped Titan's three-surface platform — public site, client CMS, and vendor portal — as one Laravel 12 + Inertia/React SSR application, live since Jul 2026

Freelance — Full-Stack Developer | Multiple Clients | 2024 – Present

- Built an e-commerce platform with Next.js 14, TypeScript (strict mode), PostgreSQL, and Prisma — 23 API endpoints, dual-role auth, running on a self-managed VPS
- Built an education platform with Laravel 11 and Vue 3 — anonymized student-donor matching with dual-key privacy architecture across 35+ database tables

EARLIER EXPERIENCE

Systems Development Engineer — BDWholeseller, Dhaka | Apr – Sep 2019
Consumer Technology Expert — David Jones, Sydney | Oct 2017 – Jan 2019
SAP Operator — Robi Axiata Ltd., Dhaka | Feb – Jul 2016

EDUCATION

B.Eng. in Mechatronics, Robotics & Automation Engineering — Monash University Malaysia | 2011 – 2016

CERTIFICATIONS

- **Agentic AI and AI Agents for Leaders** — Vanderbilt University/Coursera, 2025
- **Google Data Analytics Professional Certificate** — Google/Coursera, 2025 · **Software Engineering Specialization** — HKUST/Coursera, 2023

PUBLICATIONS

- "Democratising data insights for improved project monitoring and accountability" — 42nd WEDC International Conference, 2021